FORGE Innovation & Ventures

PRODUCT INNOVATION HYPOTHESES

TARGET CUSTOMER:

FORGE INNOVATION HANDBOOK

HYPOTHESES

No. OF CUSTOMERS VALIDATED WITH

PROBLEM

- ~$16B lost annually in India due to poor quality
- Manual inspection fails at high speeds (20–30% error rate)

0 / <5 / <10 / <20 / >=20

CONSTRAINTS & BARRIERS

- Cameras can't see all defects ("Optics Gap")
- Must integrate without slowing production
- Low trust in software-only AI

0 / <5 / <10 / <20 / >=20

VALUE / BENEFITS

- 99.9% defect detection accuracy
- Major COQ reduction & waste control
- Consistent, fatigue-free inspection

0 / <5 / <10 / <20 / >=20

SOLUTION & USABILITY

- Optics-first, patent-pending hardware
- Full-stack: Camera + AI + Automation
- Industry-ready, easy-to-use systems

0 / <5 / <10 / <20 / >=20

PRICE

- CAC: $8K | CLV: $1.4M
- CLV/CAC: 17.5
- $663K in active purchase orders

0 / <5 / <10 / <20 / >=20

Forge Innovation Rubric

FORGE
Innovation & Ventures

NECESSITY & SIGNIFICANCE	7/20	Real-world application scenarios	0 $\boxed{2}$ 4
		Wide problem incidence	0 $\boxed{2}$ 4
		High severity/impact of problem	0 $\boxed{3}$ 6
		User/Customer validated use-cases	$\boxed{0}$ 3 6
NOVELTY & ORIGINALITY	9/20	Unique & differentiated solution concept	0 $\boxed{3}$ 6
		Holistic solution offering conceptually validated functionality	0 $\boxed{2}$ 4 6
		(Futuristic) Potential to emerge as an enduring solution	0 $\boxed{4}$ 8
CUSTOMER MOTIVATION	4/20	Customers have assigned highest priority & urgency	$\boxed{0}$ 6
		Customers have failed with alternative solutions	0 $\boxed{2}$ 4
		Customers have strong intent to co-create & standardise solution	0 $\boxed{2}$ 6
		High degree of intent to invest in the immediate term	$\boxed{0}$ 4
STRONG VALUE PROPOSITION	3/20	Demonstrate functional/operational benefits	0 $\boxed{3}$ 6
		Significant gains measurable, quantified & validated	0 4 8
		Value proposition proven with reasonable number of customers	$\boxed{0}$ 6
DEMONSTRATED PROOF OF VALUE	6/20	Field tested for utility, usability, integration & deployment constraints	$\boxed{0}$ 4 6 8
		Strong indication of product adoption through successful field trails	0 $\boxed{3}$ 6
		Proven for ease-of adoption and accepted as permanent solution	0 $\boxed{3}$ 6

29 /100

 2.0

PROBLEM - CUSTOMER - PRODUCT

Getting 'em all right!

List all the beneficiaries who benefit or gain from solving this problem.

Select that beneficiary profiled in this canvas:

- ☐ ____________________
- ☐ ____________________
- ☐ ____________________
- ☐ ____________________
- ☐ ____________________

Is this Beneficiary the RIGHT Customer?

Customer-Motivation	Low	Med	High
+ faces the problem most severely, and stands to benefit the most	☐	☐	☐
+ aware of the problem and able to set clear expectations on gains/benefits	☐	☐	☐
+ dissatisfied with existing solutions as they are complex, expensive, unusable, or ineffective	☐	☐	☐

Customer-Commitment	Low	Med	High
+ willing to describe solutions attempted and specify deployment/ usability constraints	☐	☐	☐
+ strong intent to co-create an effective solution	☐	☐	☐
+ willing to define the evaluation/acceptance criteria for buying the product	☐	☐	☐
+ indicates the preferred price range for buying a product	☐	☐	☐

PV-CD Progress Indicator

No of Customers	0	<5	<10	<20	>=20
Problem definition documented as is being experienced by the beneficiary	☐	☐	☐	☐	☐
Magnitude of the problem (PAIN) has been measured and validated with beneficiary	☐	☐	☐	☐	☐
Incidence/Occurrence of the problem (FREQUENCY) has been assessed and validated with beneficiary	☐	☐	☐	☐	☐
Challenges/limitations of current solutions attempted have been discussed/analysed	☐	☐	☐	☐	☐
Gaps/Challenges with existing commercial products have been analysed/documented	☐	☐	☐	☐	☐
Adoption barriers have been discussed/documented	☐	☐	☐	☐	☐

01 Problem Description

Can you describe the problem in your own words exactly how you experience it?

We lose huge money due to unreliable quality checks. Manual inspection misses 20–30% defects at 120 BPM, damaging client trust and reputation.

02 Tried Solutions

How have you tried to solve this problem before? Can you explain the solution?

We added more inspectors and used standard AI camera software comparing parts to "good" images.

03 Gaps

Why did you not make that solution permanent? What were the gaps in the solution?

Manual checks are slow and costly. Software-only AI fails due to the Optics Gap—poor lighting and limited visibility mean defects aren't actually seen.

04 Constraints/Barriers

What are the other constraints/challenges that have to be overcome or accommodated while solving the problem?

The solution must work at high speed, give 360° internal inspection, fit existing automation, and be easy to maintain.

05 Benefits

How will you benefit from solving the problem? How much is solving this problem worth to you?

Achieving 99.9% accuracy removes COQ, saving millions. A permanent solution is worth about $1.4M CLV.

06 Importance

How important to you is solving this problem among other priorities?

Top Priority. Reducing COQ is critical for scaling and competing in global export markets.

CHALLENGE STATEMENT

Describe the challenge statement as it is faced in the real-world by the target user

1. Reduce Cost of Poor Quality (COQ) in high-speed manufacturing.
2. Achieve 99.9% inspection accuracy beyond human and standard AI limits.

GAP ANALYSIS

Alternatives

How does the target user solve the challenge today?

What are the other alternatives available in the market?

Gaps

What are the gaps in the way the challenge is solved today?

What are the gaps/shortcomings?

1. Humans miss 20–30% defects at line speed
2. Software-only AI fails due to optics limitations

CONSTRAINTS

What are the usability/ deployment constraints to be considered for the solution to be easy to use and to adopt?

1. No production slowdown; harsh factory-ready
2. Must fit existing automation systems

Usability

Skills/Expertise
Time Consumption
Physical/Personal risk
Product Training
Changes to habits/Process
Accessibility

Deployment

Total Cost of Ownership
Maintenance/Service
Installation/Integration
Resources/Materials
Time Consumption
OTHERS:

OTHERS:

USE-CASE

From the different scenarios in which the challenge is faced, describe the specific use-case you wish to target

1. Automated defect inspection at high-speed production lines
2. Achieves 99.9% quality accuracy, reducing COQ with AugurAI

TARGET USER

For this use-case describe the profile of the target user that you expect to adopt the solution

1. Plant Managers & Quality Engineers
2. Automotive, FMCG, Pharma, Aerospace sectors

UTILITY/FEATURES

Jobs

Describe the specific job the target user wants to complete (get done)?

Outcomes

What are the expected outcomes? How do you quantify them?

1. Borespekt: Internal bore inspection (5–90 mm)
2. PrIns: 360° inspection at 120 BPM by AugurAI

Pains

What are the pains to be relieved?

Gains

What are the gains to be created?

REFERENCES

Indicate other details or reference materials to know more about the user/use-case and the existing solutions/alternatives

VPC ACTIVITY GUIDE

FORGE Innovation & Ventures

Startup Name:	Date:	Version:
Team Members:		

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Pains to be relieved

Cost: What does your customer find too costly? (takes a lot of time, costs too much money, requires substantial efforts)

Feeling Bad: What makes your customer feel bad? (frustrations, annoyances, things that give them a headache)

Performance of Known Competition: How are current solutions underperforming for your customer? (lack of features, performance, malfunctioning)

Main Difficulties & Challenges: What are the main difficulties and challenges your customer encounters? (understanding how things work, difficulties getting things done, resistance)

Negative social consequences: What negative social consequences does your customer encounter or fear? (loss of face, power, trust, or status)

Risks: What risks does your customer fear? (financial, social, technical risks, or what could go awfully wrong)

Keeps Awake at Night: What's keeping your customer awake at night? (big issues, concerns, worries)

Common Mistakes: What common mistakes does your customer make? (usage mistakes)

Barriers: What barriers are keeping your customer from adopting solutions? (upfront investment costs, learning curve, resistance to change)

Customers Jobs-to-be-done

Functional Job (Perform, complete a specific task, solve a specific challenge)

1.Detect defects accurately at high production speeds without stopping the line
2.Ensure consistent product quality to meet customer and export standards

Social Job (Gain power, looking good or status)

1.Be seen as a reliable, high-quality manufacturer by clients and OEMs
2.Protect personal and company reputation through zero-defect delivery

Emotional Job (aesthetics, feel good, security)

1.Feel confident and stress-free about quality decisions
2.Avoid anxiety caused by defect escapes and customer complaints

Basic Needs (Communication, nutrition, belonging, etc)

1.Simple, dependable inspection system that works daily without complexity
2.Clear quality data and alerts for quick decision-making

Gains to be created

Savings: Which savings would make your customer happy? (time money and effort)

Outcome Expectations: What outcomes does your customer expect and what would go beyond his/her expectations? (quality level, more of something, less of something)

Current Solutions – Delight?: How do current solutions delight your customer? (specific features, performance, quality)

Life or Job – Easier?: What would make your customer's job or life easier? (flatter learning curve, more services, lower cost of ownership)

Positive Social Consequences: What positive social consequences does your customer desire? (makes them look good, increase in power, status)

What are customers looking for? (e.g. good design, guarantees, specific or more features

What do customers dream about? (e.g. big achievements, big reliefs)

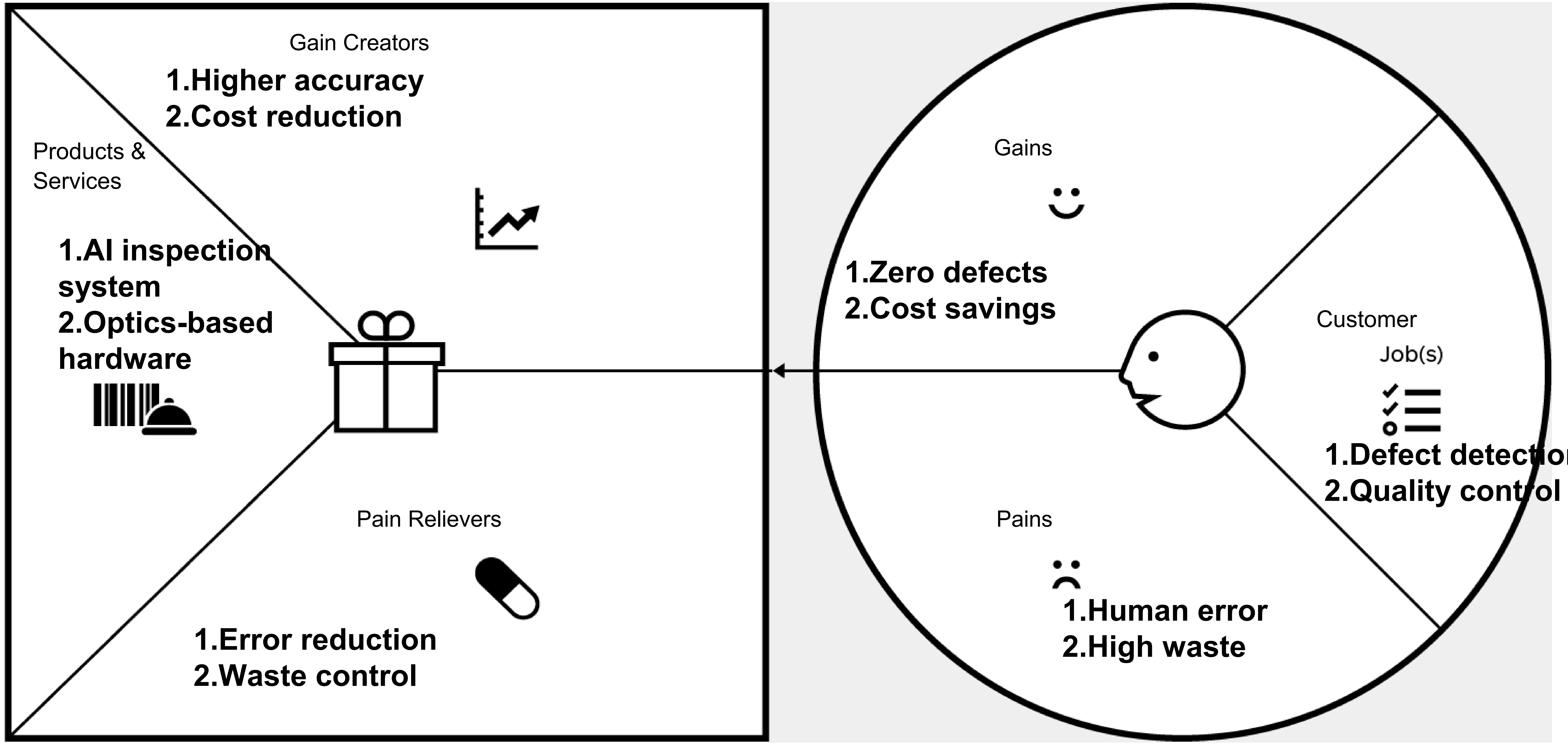
How does your customer measure success and failure? (e.g. performance, cost)

What would increase the likelihood of adopting a solution? (e.g. lower cost, less investments, lower risk, better quality, performance, design)

The Value Proposition Canvas

Value Proposition

Customer Segment



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VPC ACTIVITY GUIDE

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Our value proposition is for (Target user)

1. Quality managers
2. Plant heads

who (Challenge statement)

1. Missed defects
2. High COQ

the (Solution)

1. AI inspection
2. Optics-first system

is a (Product category/theme)

1. Industrial AI
2. Quality automation

it has (Compelling reason to buy)

1. 99.9% accuracy
2. Cost savings

unlike (Primary competitive alternatives)

1. Manual inspection
2. Software-only AI

our Product (Primary differentiation)

1. Optics-first design
2. Full-stack solution

•Multi-Camera Array (for simultaneous angles).

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MUP CONCEPTS GENERATION

FORGE INNOVATION HANDBOOK

VALUE PROPOSITION

What are your Products & Services?
What are your Gain Creators?
What are your Pain Relievers?

ADOPTION BARRIERS

What are the constrains/challenges that have to be overcome or accommodated while solving the problem?

CONCEPT LEGOS

Concept Generation

TOOLS/TECHNIQUES

FUNCTIONS	Option #1	Option #2	Option #3	Option #4	Option #5
Internal Defect Mapping	Specialized Endoscopic Lenses (for narrow access).	Robotic Probe Integration (for precise movement).	Standardized Testing Protocols.	degree Rotating Optics (to see all internal walls)	
Quality Assurance Automation	Real-time Pass/Fail Signal Triggers.	Digital Inspection Logs (for audit trails).	"Plug-and-Play" Industrial Enclosures.	Automated Calibration Software.	
High-Speed Surface Scanning	High-Speed Strobe Lighting (to freeze motion).	Multi-Camera Array (for simultaneous angles).	120 BPM Sync Sensors (to match production speed).	Edge Computing Hardware (for instant image processing).	
Invisible Flaw Detection	Shadow Mapping Techniques (to find surface dents).	Laser Deviation Sensors (to detect height changes).	Polarized Filters (to remove glare from shiny plastic).	Patent-pending Lighting Geometry.	
Human Error Elimination	Physics-based Image Enhancement.	Pre-trained Neural Networks.	Deep Learning Algorithms (to classify blowholes/scratches) .	Touch-screen Operator Interface.	

FUNCTIONS	Optics-First AI Inspection System CONCEPT #__	Full-Stack Vision + Automation Integration CONCEPT #__	Fully Autonomous Predictive Quality Intelligence CONCEPT #__
Defect detection & quality inspection **High-speed production line monitoring** **Automation integrationDecision support for quality control** **Cost of Quality (COQ) reduction**	**Capability (how feasible with current capabilities):** High — Core AI, hardware optics, and integration expertise already exist. **Usability (overcoming adoption barriers):** High — Industry-ready system, easy integration into factory workflow. **Feasibility (time & cost to build prototype):**High — Prototype already validated with strong traction.	**Capability:**Medium–High — Requires robotics + automation coordination. **Usability:**High — Reduces manual inspection and operator fatigue. **Feasibility:**Medium — Higher development and deployment effort.	**Capability:**Medium — Needs advanced predictive AI & deeper factory integration. **Usability:**Medium — Requires trust-building and operational training. **Feasibility:**Low–Medium — Higher validation, time, and cost required.
Capability How feasible is the concept to build considering your capabilities? **Usability** How usable is the prototype overcoming adoption barriers to test the value proposition? **Feasiblity** How feasible is the solution to build considering the time and cost in building the prototype?	0 - 10 Capability: 8 Usability: 8 Feasiblity: 8	0 - 10 Capability: 7 Usability: 8 Feasiblity: 6	0 - 10 Capability: 6 Usability: 7 Feasiblity: 5

MUP (Tech Canvas)

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USER/USE CASE

Manufacturing plants (bottling, automotive, FMCG, pharma) needing high-speed automated defect detection (e.g., 120 BPM production lines).

PIR: IQ:

P: High Cost of Quality (~$16B in India), 20–30% human inspection error.I: Standard cameras cannot detect critical integrated with factory lines. micro-defects (“Oonptics Gap”).R: Real-time, highly accurate (99.9%) automated inspecti

VALUE PROPOSITION:

“Optics-First” AI system delivering 99.9% defect detection, reducing waste, labor cost, and COQ while operating at full production speed

FEATURES AND FUNCTIONALITIES:

Patent-pending advanced lighting & lens systemAI-based defect detectionReal-time inspection at high speed

ACTORS AND ROLES:

Quality Manager – reviews analytics

SYSTEM CONTEXT:

Installed on production line → Captures product images → AI processes images → Sends signal to reject/accept → Data logged to dashboard.

SYSTEM DESIGN:

Camera + Optics Module → Edge Computing Unit (AI model) → PLC/Automation Interface → Dashboard (Cloud/Local Server).

UI/UX:

Simple dashboard interface

Real-time defect alerts

Visual defect highlighting

Performance & accuracy metrics display

SYSTEM CONSTRAINTS:

Must operate at high production speedLighting & environmental variabilityIntegration with legacy factory systemsIndustrial reliability requirements

INPUT:

PHYSICAL QUANTITY: | SENSOR/TRANSDUCER:

Physical Quantity: Visual defects (dents, scratches, shape deviations)Sensor/Transducer: Industrial cameras + specialized optics + lighting

INTERFACE :

Ethernet

COMPUTE:
HARD DISK MEMORY:

POWER:

AC power supply

INTERFACE :

OUTPUT

ACTUATOR : | PHYSICAL QUANTITY:

Actuator: Reject mechanism / robotic armPhysical Quantity: Accept/Reject signal + defect classification data

HMI:

Industrial dashboard (PC/Tablet) with real-time monitoring & reports.

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PRODUCT CANVAS

Startup Name:

Date:

Version:

Team Members:

FORGE INNOVATION HANDBOOK

PROBLEM STATEMENT 01

Which problem do you aim to solve?

*Describe the problem as it is faced in the real world. If you are solving a technical problem then find a real-world product/service where this tech is currently applied, that's a good starting point.

Why is solving this problem very important?

Who are the different beneficiaries?

What goal can be achieved by solving this problem?

What gaps/challenges - in the way the problem is solved today, have to be overcome?

What are the different real-world scenarios where this problem is faced most critically?

TARGET CUSTOMER 02

Which beneficiary (type/category) gains most from solving this problem?

What should we look for if we have to find real-world customers of this beneficiary (type/category) in the real-world?

What does this beneficiary (type/category) expect as benefits/outcomes?

What would it be worth for this beneficiary (type/category) to solve this problem?

PRODUCT 03

How are you solving this problem for your target customer?

What constraints faced by this customer are you overcoming with your solution?
*costs, resources, skills, operating conditions, time, convenience

How do you make this solution easy to adopt and use for your target customer?

VALUE PROPOSITION 04

Does your product help your target customer make or save money?

How is your product better than the alternative available today?

What extra benefits does the customer get from your product?

How do you make it easy for the customer to measure or experience the benefits?

UNFAIR ADVANTAGE 05

In what other ways can your customer achieve the goal?

What other offerings in the market does your target customer compare your solution with now or is likely to do so in the future?

Who are your most important present and future competitors?

What serious business advantages do you enjoy over the competitors

What serious tech, operating or business advantages do you enjoy over your current or future competitors?

CUSTOMER ACQUISITION & SALES 06

How will you create awareness and generate leads for your product?

How do you increase the conversion from leads to trials, and from trials to sales?

Where would your customers actually buy your product from?

What is the channel for distribution and delivery of your product to the customer?

PRICE & REVENUE MODEL 07

Who pays for your product? Who is the buyer & user? Are they the same?

For what value is your target customer willing to pay?

How much is your solution worth to your target customer?

Are other sources of revenue possible from other beneficiaries?

METRICS 08

Which outcomes should you measure to validate the assumptions in the creation & delivery of your core value proposition?

Are you maximising the conversion rate in the Get-Keep-Grow pipeline for your target customers?